

SSD_Perf usage reference

Flame Graph can provide visual performance analysis capabilities, which are presented in SVG format, and can be zoomed to view parts, which is convenient for quickly locating CPU performance bottlenecks. Flame Graph counts the proportion of time the code-path is executed during the sampling period. The vertical axis is the code-path, and the call stack is generally from the bottom to the top. The horizontal axis is the occupancy time of a stack. The larger the span, the higher the proportion of CPU it occupies, that is, the most CPU-intensive. The stacks are arranged in alphabetical order on the horizontal axis. The color depth is only for distinction, not special. significance.



The following will describe the Flame Graph generation process:

1. Get system execution status with perf

- Step 1: Execute perf record on EVB

```
# . / perf record -a -g sleep 10

-a for cpu

-o Specify the output file , if not specified, it will be
automatically saved in the current directory and named perf
.data

-g record call stack

sleep specifies the duration in seconds
```

If it is displayed during execution, it means that the system lacks GCC runtime library, please put libgcc_s.so.1 in /mnt or other paths, and add the path to LD_LIBRARY_PATH

```
# export LD_LIBRARY_PATH=/mnt:/lib
```

- Step 2: Execute perf report to expand the data just saved for analysis

```
# ./perf report
```

The call stack and running time ratio of each leaf function will be displayed directly on the console

To perform FlameGraph analysis directly, perf report is not necessary

2. Generate a flame graph with data

- Step 1: Execute perf script on EVB to parse the data, and archive the output to p1

```
# ./perf script -i perf.data > p1
```

- Step 2: Switch to the PC side, [click to get](https://github.com/brendangregg/FlameGraph) [https://github.com/brendangregg/FlameGraph] FlameGraph, and execute the stackcollapse-perf command provided by FlameGraph on the PC side to fold p1 and save it to disk

```
# stackcollapse-perf.pl p1 &> perf.folder
```

- Step 3: Execute p1 provided by FlameGraph on the PC side to convert perf.folder to svg file. The generated svg file can be opened directly with the browser (Chrome or Firefox is recommended)

```
# flamegraph.pl perf.folder > perf.svg
```

Generate flame graph (Flame Graph) command:

```
./perf record -a -g -F 2599 -p 451 sleep 30  
./perf script -i perf.data > p1  
./stackcollapse-perf.pl p1 &> perf.folder  
./flamegraph.pl perf.folder > perf.svg
```

refer to:

<https://www.jianshu.com/p/9a597ff4f5de>

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<https://linux.cn/article-4670-1.html> [<https://linux.cn/article-4670-1.html>]